

# **SQL-Based Data Analysis Using Filtering, Aggregation and Business Queries**

## **1. INTRODUCTION**

This project focuses on analyzing the Superstore sales dataset using SQL. The objective is to explore the dataset, apply filtering conditions, perform aggregations, and answer business-related questions using SQL queries. The analysis helps in understanding sales performance, customer behavior, product trends, and data quality.

## **2. DATASET OVERVIEW**

The Superstore dataset contains information related to customer orders, products, sales, profit, quantity, categories, regions, and order dates.

Key attributes used in the analysis include:

- Order ID
- Order Date
- Customer Name
- Product Name
- Category
- Sub-Category
- Region
- State
- Sales
- Quantity
- Profit

The dataset was imported into MySQL Workbench and stored in the table named superstore.

## **3. DATA EXPLORATION**

The first step was to understand the structure of the dataset.

The following operations were performed:

- Displayed the first 10 rows of the dataset.
- Counted the total number of records.
- Examined the schema and column names.

## Observations

- The dataset contains sales transactions from different regions.
- Multiple product categories and sub-categories are available.
- Sales and profit values are recorded for each order.

|             |              |         |                    |
|-------------|--------------|---------|--------------------|
| Result Grid | Filter Rows: | Export: | Wrap Cell Content: |
| total_rows  |              |         |                    |
| 9694        |              |         |                    |

## 4. DATA FILTERING

Several filtering queries were executed to analyze specific portions of the data.

The following filters were applied:

- Orders from the West region.
- Orders belonging to the Furniture category.
- Sales greater than 1000.
- Profit greater than 500.
- Orders placed during 2017.

## Purpose

Filtering helps focus on specific business scenarios and allows detailed analysis of selected records.

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Customer ID | Customer Name     | Segment     | Country       | Sales    | City          | State          | Postal Code | Region  |
|--------|----------------|------------|------------|----------------|-------------|-------------------|-------------|---------------|----------|---------------|----------------|-------------|---------|
| 1614   | CA-2017-138380 | 12/21/2017 | 12/25/2017 | Standard Class | YS-21880    | Yana Sorensen     | Corporate   | United States | 1000.02  | Oakland       | California     | 94601       | West    |
| 6211   | CA-2017-119809 | 8/18/2017  | 8/25/2017  | Standard Class | YS-21880    | Yana Sorensen     | Corporate   | United States | 1000.02  | Seattle       | Washington     | 98103       | West    |
| 5157   | CA-2017-163006 | 6/30/2017  | 7/4/2017   | Second Class   | GH-14410    | Gary Hansen       | Home Office | United States | 1001.584 | Chicago       | Illinois       | 60653       | Central |
| 6908   | CA-2016-140249 | 9/27/2016  | 10/3/2016  | Standard Class | SW-20455    | Shaun Weien       | Consumer    | United States | 1001.584 | Seattle       | Washington     | 98103       | West    |
| 2482   | CA-2017-147956 | 12/24/2017 | 12/31/2017 | Standard Class | AH-10210    | Alan Hwang        | Consumer    | United States | 1003.62  | Seattle       | Washington     | 98105       | West    |
| 409    | CA-2017-117457 | 12/8/2017  | 12/12/2017 | Standard Class | KH-16510    | Keith Herrera     | Consumer    | United States | 1004.024 | San Francisco | California     | 94110       | West    |
| 1506   | CA-2015-153388 | 8/1/2015   | 8/7/2015   | Standard Class | PC-19000    | Pauline Chand     | Home Office | United States | 1004.976 | Los Angeles   | California     | 90004       | West    |
| 1631   | CA-2016-106530 | 5/8/2016   | 5/8/2016   | Same Day       | CL-12565    | Clay Ludtke       | Consumer    | United States | 1006.056 | Cleveland     | Ohio           | 44105       | East    |
| 949    | US-2017-110576 | 11/28/2017 | 12/2/2017  | Standard Class | RB-19795    | Ross Baird        | Home Office | United States | 1007.232 | Philadelphia  | Pennsylvania   | 19120       | East    |
| 6579   | CA-2014-103660 | 8/25/2014  | 8/30/2014  | Standard Class | MW-18220    | Mitch Webber      | Consumer    | United States | 1007.944 | Seattle       | Washington     | 98103       | West    |
| 437    | CA-2016-147375 | 6/12/2016  | 6/14/2016  | Second Class   | PO-19180    | Philisse Overcash | Home Office | United States | 1007.979 | Chicago       | Illinois       | 60623       | Central |
| 7082   | CA-2016-106950 | 9/2/2016   | 9/6/2016   | Standard Class | JE-15715    | Joe Elijah        | Consumer    | United States | 1012.68  | Charlotte     | North Carolina | 28205       | South   |

## 5. AGGREGATION ANALYSIS

Aggregate functions were used to summarize the dataset.

Functions used:

- SUM()
- AVG()
- MAX()

- MIN()

Metrics calculated:

- Total Sales
- Total Quantity Sold
- Average Sales
- Highest Sale
- Lowest Sale
- Total Profit

### Observations

- Aggregate functions provide a quick summary of overall business performance.
- Sales and profit metrics help measure revenue generation and profitability.

|                      |              |         |                    |
|----------------------|--------------|---------|--------------------|
| Result Grid          | Filter Rows: | Export: | Wrap Cell Content: |
| total_sales          |              |         |                    |
| ▶ 2272449.8562999545 |              |         |                    |

|                |              |         |                    |
|----------------|--------------|---------|--------------------|
| Result Grid    | Filter Rows: | Export: | Wrap Cell Content: |
| total_quantity |              |         |                    |
| ▶ 36749        |              |         |                    |

## 6. GROUP BY ANALYSIS

Grouping was used to compare performance across different categories.

Analysis performed:

- Sales by Region
- Profit by Region
- Sales by Category
- Quantity Sold by Sub-Category

### Observations

- Some regions contribute more to sales than others.
- Product categories show different revenue patterns.
- Certain sub-categories generate higher sales volumes.

| Result Grid |         |             | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------|-------------|--------------|---------|--------------------|
|             | region  | total_sales |              |         |                    |
| ▶           | West    | 713471.34   |              |         |                    |
|             | East    | 672194.05   |              |         |                    |
|             | Central | 497800.87   |              |         |                    |
|             | South   | 388983.59   |              |         |                    |

| Result Grid |         |              | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------|--------------|--------------|---------|--------------------|
|             | region  | total_profit |              |         |                    |
| ▶           | West    | 106021.15    |              |         |                    |
|             | East    | 90672.01     |              |         |                    |
|             | South   | 46035.69     |              |         |                    |
|             | Central | 40128.9      |              |         |                    |

## **7. TOP PERFORMING PRODUCTS AND CUSTOMERS**

Sorting and limiting operations were used to identify high-performing entities.

The following analyses were conducted:

- Top 10 Products by Sales
- Top 10 Customers by Sales
- Top Categories by Sales

### **Observations**

- A small number of products contribute significantly to total sales.
- Top customers account for a considerable share of revenue.

| Result Grid |                                                    |             | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: |
|-------------|----------------------------------------------------|-------------|--------------|---------|--------------------|-------------|
|             | Product Name                                       | total_sales |              |         |                    |             |
| ▶           | Canon imageCLASS 2200 Advanced Copier              | 61599.82    |              |         |                    |             |
|             | Fellowes PB500 Electric Punch Plastic Comb Bind... | 27453.38    |              |         |                    |             |
|             | Cisco TelePresence System EX90 Videoconferen...    | 22638.48    |              |         |                    |             |
|             | HON 5400 Series Task Chairs for Big and Tall       | 21870.58    |              |         |                    |             |
|             | GBC DocuBind TL300 Electric Binding System         | 19823.48    |              |         |                    |             |
|             | GBC Ibimaster 500 Manual ProClick Binding System   | 19024.5     |              |         |                    |             |
|             | Hewlett Packard LaserJet 3310 Copier               | 18839.69    |              |         |                    |             |
|             | HP Designjet T520 Inkjet Large Format Printer ~... | 18374.9     |              |         |                    |             |
|             | GBC DocuBind P400 Electric Binding System          | 17965.07    |              |         |                    |             |
|             | High Speed Automatic Electric Letter Opener        | 17030.31    |              |         |                    |             |

## **8. BUSINESS USE CASES**

Business-oriented queries were executed to generate useful insights.

The following analyses were performed:

- Monthly Sales Trend
- Monthly Profit Trend

- Top 5 Customers
- Most Profitable Products
- Top States by Sales

### Observations

- Monthly trends help identify seasonal patterns.
- Profitability varies across products.
- Certain states contribute significantly to total sales.

|             |             |              |  |         |  |                    |  |
|-------------|-------------|--------------|--|---------|--|--------------------|--|
| Result Grid |             | Filter Rows: |  | Export: |  | Wrap Cell Content: |  |
| month       | total_sales |              |  |         |  |                    |  |
| NULL        | 2272449.86  |              |  |         |  |                    |  |

## 9. DATA QUALITY VALIDATION

Data quality checks were performed to ensure the reliability of the dataset.

Checks included:

- Duplicate Order IDs
- Missing Customer Names
- Missing Product Names
- Negative Sales Values
- Negative Profit Values

### Observations

- No missing customer records were found.
- No missing product records were found.
- Duplicate records were identified and reviewed where applicable.
- Data quality was generally satisfactory for analysis.

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

|  |            |                |                     |                    |             |
|--|------------|----------------|---------------------|--------------------|-------------|
|  | total_rows | order_id_count | customer_name_count | product_name_count | sales_count |
|  | 9694       | 9694           | 9694                | 9694               | 9694        |

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

|        |          |            |           |           |             |               |         |         |      |       |             |        |            |          |              |              |
|--------|----------|------------|-----------|-----------|-------------|---------------|---------|---------|------|-------|-------------|--------|------------|----------|--------------|--------------|
| Row ID | Order ID | Order Date | Ship Date | Ship Mode | Customer ID | Customer Name | Segment | Country | City | State | Postal Code | Region | Product ID | Category | Sub-Category | Product Name |
|--------|----------|------------|-----------|-----------|-------------|---------------|---------|---------|------|-------|-------------|--------|------------|----------|--------------|--------------|

## 10. KEY INSIGHTS

1. Regional sales analysis shows differences in revenue contribution across regions.
2. Product categories contribute differently to overall sales performance.
3. Top customers generate a significant portion of total sales.
4. Monthly sales trends provide visibility into business performance over time.
5. Data quality checks confirmed that the dataset is suitable for analysis.

## **11. CONCLUSION**

This project demonstrated the use of SQL for data exploration, filtering, aggregation, grouping, sorting, and business analysis. The Superstore dataset was successfully analyzed to extract meaningful insights related to sales performance, customer behavior, product performance, and overall business trends. SQL proved to be an effective tool for transforming raw transactional data into actionable information.